

ANAS ESSAM MAHMOUD

ANGULAR FRONTEND DEVELOPER

El-Sheikh Zayed City, Giza, Egypt · +20 109 420 1827 · anasessam211@gmail.com

linkedin.com/in/anas-essam211 · github.com/anasessam12

SUMMARY

Angular Frontend Developer specializing in bilingual Arabic/English web applications across government, ERP/HR SaaS, ecommerce, and corporate website domains. Proven record of shipping production apps end-to-end — from admin and client portals with single sign-on (SSO) to an offline-first Electron desktop app — along with reusable published libraries and a company design system. A dedicated, fast learner and committed teammate who works closely with backend engineers on API contracts, UI/UX designers on pixel-accurate right-to-left (RTL) interfaces, and QA to ship polished, accessible features.

PROFESSIONAL EXPERIENCE

Frontend Developer — Cloud4Rain

Dec 2024 – Present · Cairo, Egypt

Sole developer on the APA government platform; also delivered feature libraries for the C4R SaaS HR suite and a publishable UI kit.

- Architected and shipped Angular apps end-to-end (admin, client, SSO/permissions, Electron voting desktop) with layered core/features/shared architecture, full role-based access control (RBAC), and bilingual Arabic/English RTL support.
- Published versioned `@cloud4rain_c4r/*` Angular libraries (ng-packagr → private npm) and built a themeable design system (`@cloud4rain/ui`) with Storybook, design tokens, and AXE accessibility checks.
- Designed an offline-first Electron voting product (Nx monorepo, IPC caches, queued vote sync with retry) and packaged installers for Windows, macOS, and Linux via electron-builder with GitHub Actions / Nx Cloud CI.
- Contributed Time Attendance features inside a team Angular 20 + Nx SaaS codebase (OnPush, NgRx, BaseHttpService, Jest), following shared conventions and Docker/Helm GitOps delivery.

Freelance Angular Developer

Jan 2024 – Present · Cairo, Egypt

- Delivered client Angular apps with CRUD flows, refresh-token authentication, responsive PrimeNG/Material UI, and performance-focused frontend architecture (see Rafrat and EGO under Projects).

Frontend Developer — Project-EGY

Dec 2022 – Nov 2023 · Cairo, Egypt

- Built reusable Angular components, pipes, interceptors, and standalone modules for responsive company and ecommerce products; improved reusability across portfolio and classified-ads platforms.
- Started as an intern (Dec 2022 – Jan 2023) building singleton services and dynamic carousel content, then progressed to full frontend development through Nov 2023.

PROJECTS

APA Electronic Voting Desktop App — Cloud4Rain · Angular 21 + Nx + Electron 41

Offline-first desktop app that lets APA committees run elections and cast votes without a network connection, then syncs results safely when back online. One Nx monorepo runs the same codebase as a web SPA and an Electron desktop app, with queued vote sync (retry + progress tracking), local IPC caches, printed PDF vote receipts, and installers for Windows, macOS, and Linux.

Highlights: *offline-first architecture and data-sync design, Electron security model (contextIsolation, typed preload bridge), Nx library boundaries, CI with GitHub Actions + Nx Cloud, and test coverage with Vitest, Jest, and Playwright.*

APA Government Web Platform — Admin, Client & SSO Portals · Cloud4Rain · Angular 17/18

Digital platform for the Egyptian Administrative Prosecution Authority, built and owned end-to-end as three portals: an internal admin portal for employees (RBAC with `hasPermission` directive, SignalR realtime, NgRx Signal Store, TanStack Query caching, in-browser ML image moderation, reusable base-table system), a public client portal where citizens submit 18 types of government e-services rendered by one config-driven form engine (~69 lazy routes), and a server-side rendered (SSR, Express) SSO portal centralizing login, users, groups, and permissions with captcha-protected auth.

Highlights: *layered architecture that scales across multiple apps, enterprise security (JWT cookies, CSRF interceptors, chunked token storage), SSR/SEO, IndexedDB draft persistence, and generic config-driven components.*

Staff Affairs & Election Cycles Libraries — @cloud4rain_c4r/* · Angular 17

Publishable Angular libraries in a multi-package npm workspace, consumed by the APA portals. Staff Affairs digitizes HR workflows: an 8-step member-registration wizard, leaves, promotions with ministerial decrees, transfers, and technical offices. Election Cycles covers the full election lifecycle: a 9-step cycle-creation wizard, committee/district hierarchy, candidates, voter access codes with PDF reports, and a live monitoring dashboard.

Highlights: *full lifecycle of publishable libraries (ng-packagr, versioned releases, npm publishing), clean public API design, CSS isolation with prefixed Tailwind builds, and complex flows broken into per-step NgRx Store/Effects state.*

C4R UI Kit — @cloud4rain/ui · Angular 21

Company design-system library: themeable components (buttons, inputs, table, dialog, dropdowns, paginator) documented in Storybook 10 with docs and a11y (AXE) addons, per-component CSS design tokens, Jest specs, and verification against Angular 17/18 host apps before publishing via Verdaccio.

Highlights: *design-token theming, reusable component API design, AXE accessibility testing, and cross-version Angular compatibility.*

Time Attendance — C4R SaaS · Angular 20 + Nx

Feature library inside a team-owned multi-app SaaS HR monorepo: shift management (including Ramadan shifts), shift assignment, and attendance correction/reporting, with six route-level lazy NgRx feature stores, a shared BaseHttpService API layer, and Jest specs alongside every component.

Highlights: *collaboration inside strict team conventions (OnPush, signal-based components, code review), Docker + Helm GitOps delivery, and Arabic-first product depth — digit normalization directives, localized date pipes, and full RTL layouts.*

Earlier Projects — Rafrat, EGO & Project-EGY · Angular · PrimeNG / Angular Material

Rafrat: responsive ecommerce platform for car spare parts in Saudi Arabia (freelance). **EGO:** classified-ads platform with filtering and pagination over large, list-heavy datasets (freelance). **Project-EGY:** dynamic company portfolio with a PMO system for project workflows and real-time progress tracking.

SKILLS

Core: Angular 17–21 · TypeScript · JavaScript (ES6+) · HTML5 · CSS3 / SCSS · RxJS · Reactive Forms · REST APIs · OOP · SOLID · Design Patterns

Architecture & State: Layered Core/Features/Shared Architecture · Standalone Components · Signals · NgRx Store/Effects · NgRx Signal Store · TanStack Query · Lazy Loading · Interceptors · Guards · Nx Monorepos · Publishable Libraries · IndexedDB · Angular SSR (Express) · SEO Optimization · Selective Preloading

Auth & Security: SSO · JWT Cookies · RBAC/Permissions (*hasPermission* directive) · CSRF & Security-Header Interceptors · Chunked Token Storage · Captcha · Secure Electron (contextIsolation, typed preload)

UI & Design Systems: PrimeNG · Angular Material · Tailwind CSS 3/4 (incl. prefixed CSS isolation) · Storybook 10 · Design Tokens · Config-Driven Dynamic Forms · Reusable Base-Table Systems · Accessibility Testing (AXE)

i18n: ngx-translate · Arabic/English Bilingual Apps · RTL Layouts · Arabic Digit Normalization · Localized Date/Time Pipes

Realtime, Desktop & ML: SignalR · Electron · Offline-First Sync (queued retry) · IPC · electron-builder (Windows/macOS/Linux) · PDF Generation (printToPDF, reports/receipts) · In-Browser ML Image Moderation (TensorFlow.js/NSFWJS)

Testing, CI/CD & Delivery: Vitest · Jest · Playwright · Karma/Jasmine · ESLint · Prettier · Git/GitHub · GitHub Actions · Nx Cloud CI · Docker · Helm/GitOps · Verdaccio · npm Publishing (ng-packagr, versioned releases)

EDUCATION

Bachelor of Computer Science — Faculty of Computers and Artificial Intelligence, Helwan University 2019 – 2023
GPA: 3.0 (Very Good) · Graduation Project: Arabic OCR with Post-Correction using AI

LANGUAGES

Arabic — Native · **English** — Professional working proficiency